

Course Schedule

Three days, two hours each. One printable page per day so it can sit next to the laptop or go on the wall. Times are minutes elapsed from the start of that day's session, not clock times — shift the whole column to fit your actual start time.

■ teaching / discussion ■ lab — hands-on in Colab ■ break

BEFORE DAY 1

Send the **Day 0 setup guide** and access-check notebook at least 5 days ahead, and the **pre-test link** alongside it. Students should arrive with a working Earth Engine project — Day 1 has no slack built in to fix registration problems mid-session.

Day 1 — Foundations + Land Cover

Slides: [day1.pdf](#) (or [day1-th.pdf](#))

Labs: [lab1_first_map](#) · [lab2_landcover](#)

0:00	Pre-test 20 MCQ + 5 Likert self-efficacy items. Not graded — tell students so before they start.	10 min
0:10	Remote sensing fundamentals Why RS at all (fieldwork limits, revisit + archive) · pixel, band, spectral signature · NDVI · the index family (EVI saturation fix, NDWI, NBR) · where the field is going.	40 min
0:50	What Earth Engine actually is Catalog + rented compute. Colab tour, the auth cell, why scale costs money.	20 min
1:10	Vibe-coding method The PROMPT recipe, the error loop, the sniff test, the anti-hallucination drill.	15 min
1:25	LAB Sentinel-2 composite + NDVI Cloud-free true-colour image of your own district, turned into a greenness map.	20 min
1:45	LAB Land cover → hectares ESA WorldCover / Dynamic World → area per class. The WorldCover-vs-Dynamic-World disagreement.	15 min
2:00	Wrap + homework Folds into the end of Lab 2. Re-run Labs 1–2 on a home province — bring the mean NDVI and tree-cover hectares to Day 2.	—

DAY 1 TIMING WAS REBALANCED TO FIT AN EXPANDED RS SECTION

The remote sensing block grew from 25 to 40 minutes to cover material adapted from *Applied Ecology* (fieldwork motivation, the EVI/NDWI/NBR index family, and current trends). The GEE model and vibe-coding blocks were tightened by 5 minutes each to make room — move a little faster through those two than the slide count suggests, and let the wrap-up fold into the tail of Lab 2 rather than standing as its own slide.

Day 2 — Forest Type + Carbon

Slides: [day2.pdf](#) (or [day2-th.pdf](#))
Labs: [lab3_forest_type_rf](#) · [lab4_carbon](#)

0:00	Recap Three quick questions on Day 1: NDVI bounds, why scale=100, why the two land-cover products disagreed.	10 min
0:10	Supervised classification concept Training points → Random Forest → a map. What "never test on training data" means and why.	20 min
0:30	LAB Random Forest forest-type map Evergreen vs deciduous from NDVI seasonal amplitude. Confusion matrix, accuracy, kappa.	45 min
1:15	BREAK Break Lab 3 is the likeliest overrun — protect this break by having a pre-made training-point fallback ready.	—
1:15	The carbon chain Biomass → carbon ($\times 0.47$) → CO ₂ e ($\times 44/12$), and where each conversion actually happens.	15 min
1:30	LAB Carbon stock: ORNL + GEDI Total t C and t CO ₂ e for the AOI. The unit trap: ORNL is already carbon, GEDI is not.	25 min
1:55	Wrap + homework Re-run Lab 3 with numPoints=50 instead of 300 and note what happens to accuracy.	5 min

Day 3 — Fire + Climate + Mini-project

Slides: [day3.pdf](#) (or [day3-th.pdf](#))

Labs: [lab5_fire](#) · [lab6_climate](#) · [miniproject_template](#)

0:00	Recap Two days of tools in one slide — what carries over is the checking habit, not the dataset IDs.	10 min
0:10	LAB Fire: FIRMS + MCD64A1 + dNBR Active fire vs burned area vs severity. The naive-dNBR-over-deciduous-forest 24× error and its fix.	40 min
0:50	LAB Climate: CHIRPS + ERA5 Monthly rainfall and temperature. The chart that explains why the fire season falls where it does.	30 min
1:20	BREAK Break Short break before the open-ended work — the mini-project needs a clear head, not momentum.	—
1:25	LAB Mini-project One province, one map, one defensible number — student's choice of topic, any lab as the template.	25 min
1:50	Post-test Identical instrument to Day 1. Answer honestly — this measures the course, not the student.	10 min
2:00	Share-outs Three minutes each: show the map, say the number, say where it came from.	—

WHERE TO FIND MORE DETAIL

This page is the timetable. For the full teaching content behind each block, see the slide deck for that day; for the lab code, see [02-notebooks/instructor/](#) ; for what to do when something breaks mid-session, see [Troubleshooting.pdf](#) .